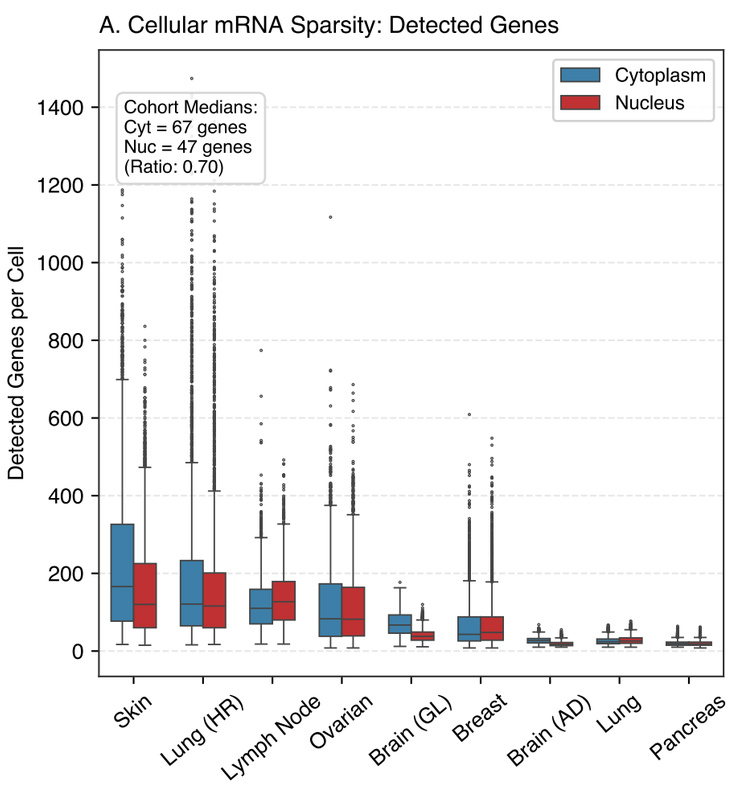
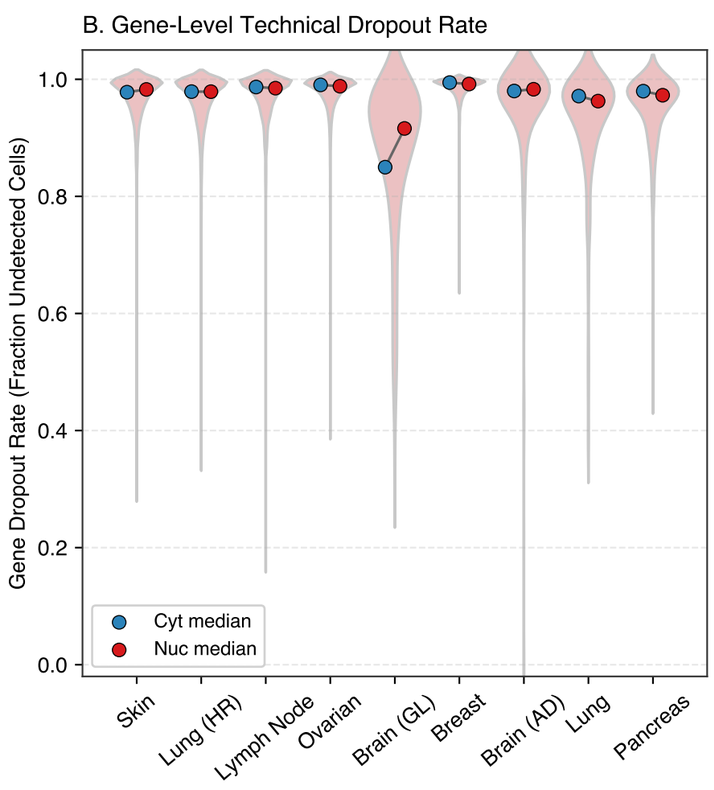
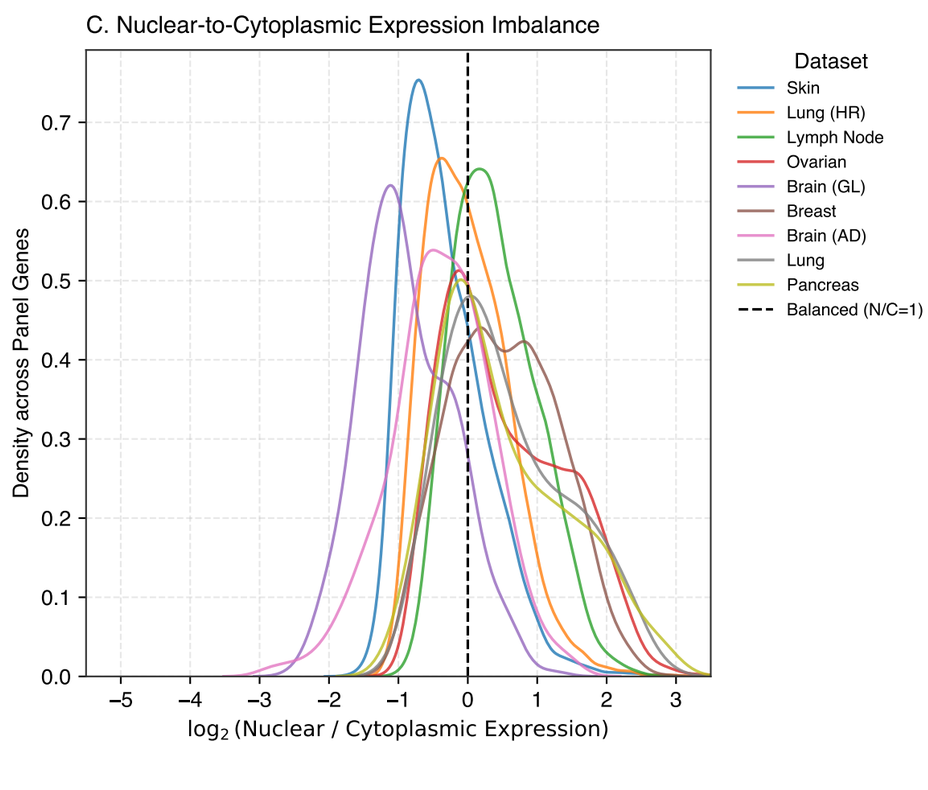
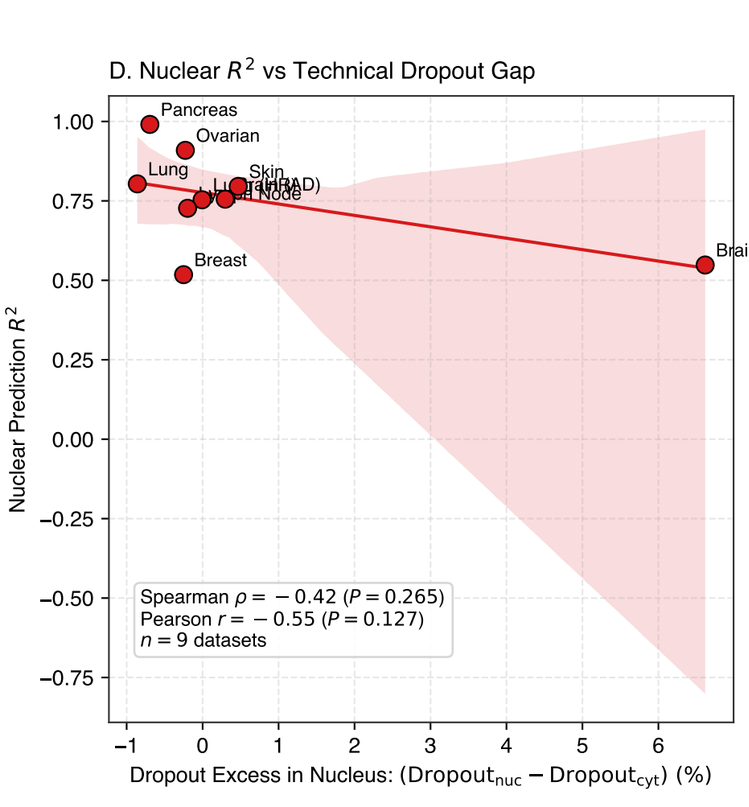
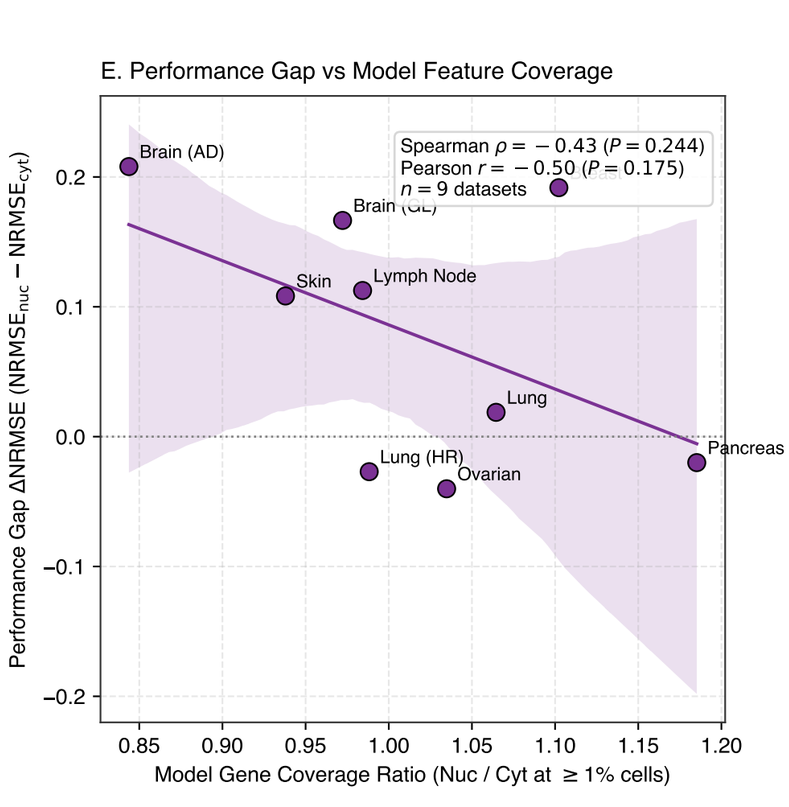
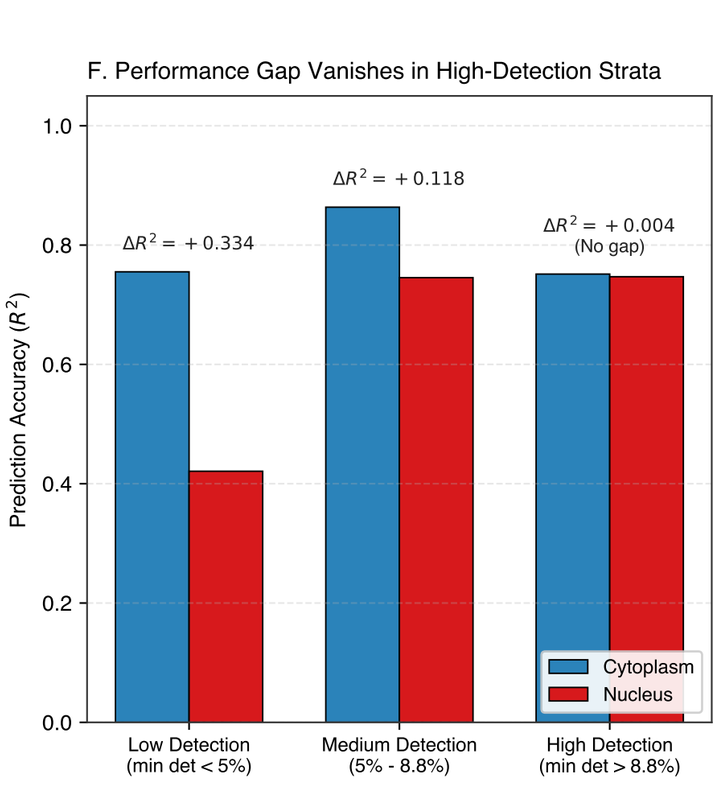

A. Cellular mRNA Sparsity: Detected Genes
Cytoplasm
Nucleus
Cohort Medians:
Cyt = 67 genes
Nuc = 47 genes
(Ratio: 0.70)
Detected Genes per Cell
0
200
400
600
800
1000
1200
1400
Skin
Lung (HR)
Lymph Node
Ovarian
Brain (GL)
Breast
Brain (AD)
Lung
Pancreas


B. Gene-Level Technical Dropout Rate
Gene Dropout Rate (Fraction Undetected Cells)
0.0
0.2
0.4
0.6
0.8
1.0
Cyt median
Nuc median
Skin
Lung (HR)
Lymph Node
Ovarian
Brain (GL)
Breast
Brain (AD)
Lung
Pancreas


C. Nuclear-to-Cytoplasmic Expression Imbalance
Density across Panel Genes
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
−5
−4
−3
−2
−1
0
1
2
3
log2 (Nuclear / Cytoplasmic Expression)
Dataset
Skin
Lung (HR)
Lymph Node
Ovarian
Brain (GL)
Breast
Brain (AD)
Lung
Pancreas
Balanced (N/C=1)


D. Nuclear R^2 vs Technical Dropout Gap
Nuclear Prediction R^2
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Pancreas
Ovarian
Lung
Skin
Breast
Brain (GL)
Spearman $\rho = -0.42$ ($P = 0.265$)
Pearson $r = -0.55$ ($P = 0.127$)
$n = 9$ datasets
−1
0
1
2
3
4
5
6
Dropout Excess in Nucleus: ($Dropout_{nuc} - Dropout_{cyt}$) (%)


E. Performance Gap vs Model Feature Coverage
Performance Gap ΔNRMSE (NRMSE$_{nuc}$ − NRMSE$_{cyt}$)
0.2
0.1
0.0
−0.1
−0.2
Brain (AD)
Brain (GL)
Skin
Lymph Node
Lung
Lung (HR)
Ovarian
Pancreas
Spearman $\rho = -0.43$ ($P = 0.244$)
Pearson $r = -0.50$ ($P = 0.175$)
$n = 9$ datasets
0.85
0.90
0.95
1.00
1.05
1.10
1.15
1.20
Model Gene Coverage Ratio (Nuc / Cyt at ≥ 1% cells)


F. Performance Gap Vanishes in High-Detection Strata
Prediction Accuracy (R^2)
0.0
0.2
0.4
0.6
0.8
1.0
$\Delta R^2 = +0.334$
$\Delta R^2 = +0.118$
$\Delta R^2 = +0.004$
(No gap)
Cytoplasm
Nucleus
Low Detection
(min det < 5%)
Medium Detection
(5% - 8.8%)
High Detection
(min det > 8.8%)